

C05 AT1 LEVEL 3



Secure Exit

SIMATS Online Programming Lab
C++ PROGRAMMING

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QUESTIONS

LEVEL 3 — HARD

Q1
recursive

Q2
functionsandpointers

2 / 2 saved

Level 3 — Hard Fibonacci (H) — recursive

✓ Submitted

Develop a recursive function to determine the Nth Fibonacci number.

Sample Input:
Enter the value of N: 7

Sample Output:
7th Fibonacci number: 13

Test Cases:
1.5
2.10
3.15
4.20

Your Code

```
1 #include<iostream>
2 using namespace std;
3 int fibonacci(int n){
4     if(n==0)
5         return 0;
6     if(n==1)
7         return 1;
8     return fibonacci(n-1)+fibonacci(n-2);
9 }
10 int main(){
11     int n;
12     cin>>n;
13     cout<<n<<"th fibonacci number: "<<fibonacci(n);
14     return 0;
15 }
16
```

Output

```
7th fibonacci number: 13
```



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Level 3 — Hard Transpose (H) — functionsandpointers

✓ Submitted

Create a function void transposeMatrix(int *matrix, int *transpose, int rows, int cols) that calculates the transpose of a provided matrix utilizing pointers.

Sample Input:
Enter rows and columns of the matrix: 2 3
Enter elements of the matrix:
1 2 3
4 5 6

Sample Output:
Transposed matrix is:
1 4
2 5
3 6

Your Code

```
1 #include<iostream>
2 using namespace std;
3 void transposeMatrix(int matrix[][10],int transpose[][10],
4 {
5     for(int i=0;i<rows;i++){
6         for(int j=0;j<cols;j++){
7             transpose[j][i]=matrix[i][j];
8         }
9     }
10 }
11 int main(){
12     int rows,cols;
13     cin>>rows>>cols;
14     int matrix[10][10],transpose[10][10];
15     for(int i=0;i<rows;i++)
16         for(int j=0;j<cols;j++)
17         cin>>matrix[i][j];
```

Output

```
Transposed matrix is :
1 4
2 5
3 6
```

